Campus Lost & Found Live Desk

JavaFX GUI Project - TCP Socket Client/Server Application

1. Project Idea

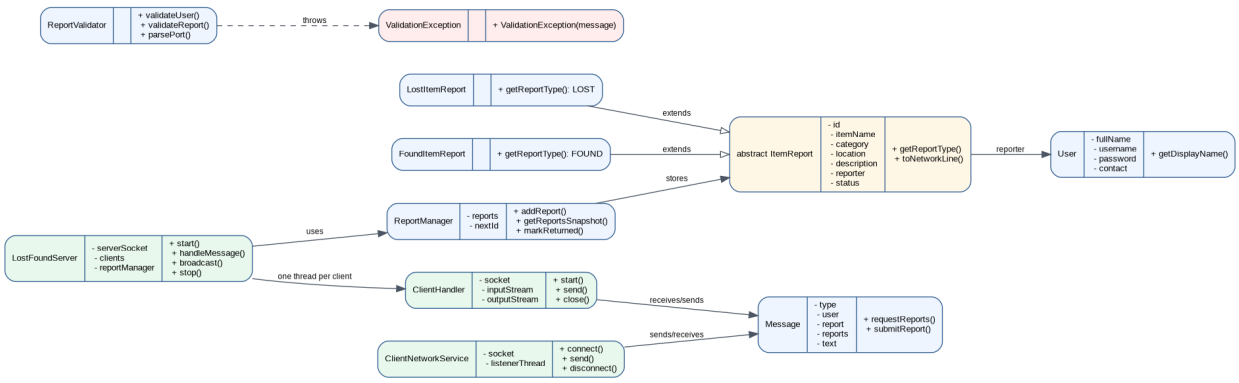
Campus Lost & Found Live Desk helps students report lost and found items on campus. Multiple student clients connect to a live admin server through TCP sockets. Students can submit reports, browse live reports, and receive admin announcements.

2. Architecture

The system is split into two separate running JavaFX programs: a Student Client and an Admin Server. The Admin Server opens a ServerSocket. Every connected client is handled by its own ClientHandler thread. Client and server exchange serialized Message objects using ObjectInputStream and ObjectOutputStream.

Component	Responsibility
Client JavaFX app	Login/register, connect to server, submit reports, browse reports.
Admin JavaFX app	Start/stop server, view clients, view reports, broadcast messages.
LostFoundServer	Accept TCP clients, validate reports, broadcast updates.
ClientHandler	Dedicated server-side thread for one client.
ClientNetworkService	Client socket connection and background listener thread.
ReportManager	Stores reports and marks reports as returned.

3. UML Class Diagram

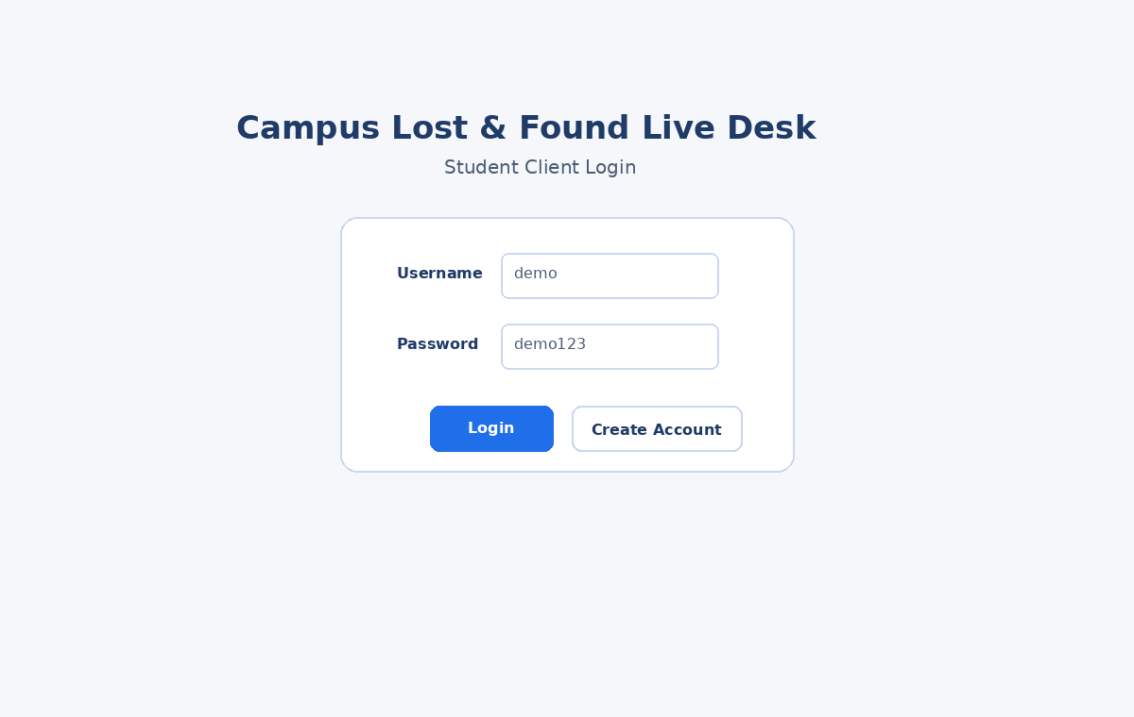


4. Requirements Coverage

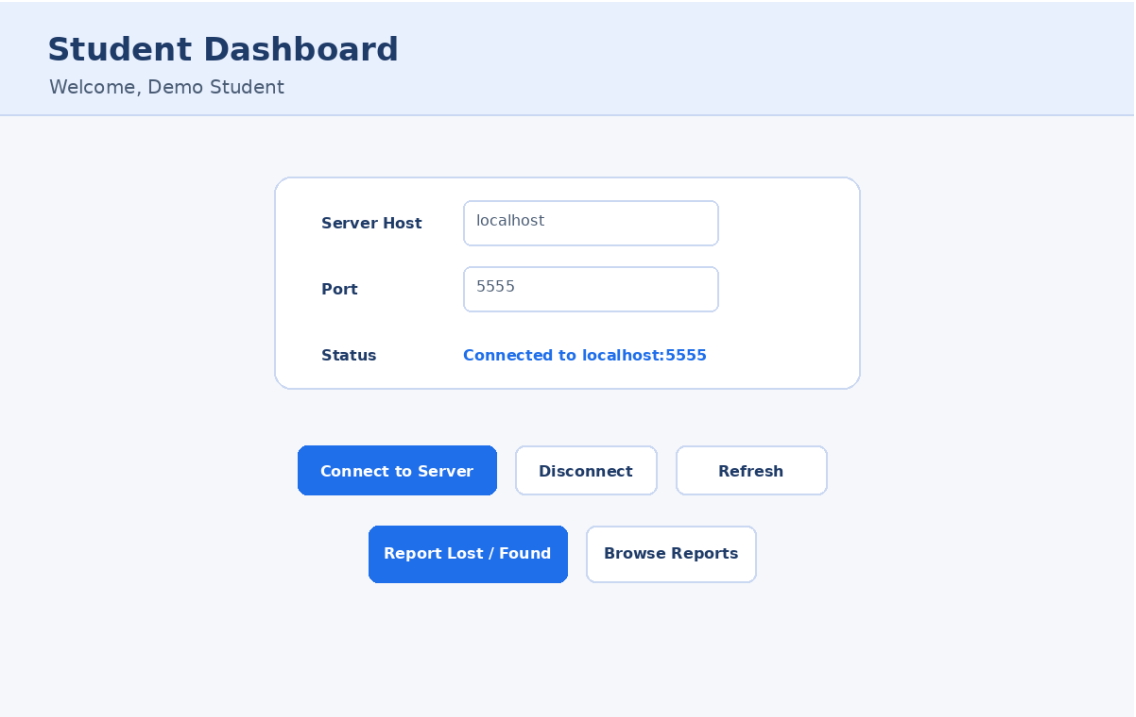
Requirement	Covered by
Encapsulation	Private model fields with getters/setters in User and ItemReport.
Polymorphism	LostItemReport and FoundItemReport override getReportType().
Exceptions	ValidationException for empty fields, invalid ports, and disconnected submissions.
JavaFX components	FXML uses TableView, ListView, ComboBox, TextArea, TextField, Button, Label, layouts.
GUI events	Controllers handle login, register, connect, submit, refresh, broadcast, mark returned.
Multithreading	Server creates one ClientHandler thread per client; client listens in a background thread.
Networking	TCP sockets between separate client and server programs.

5. Screenshots

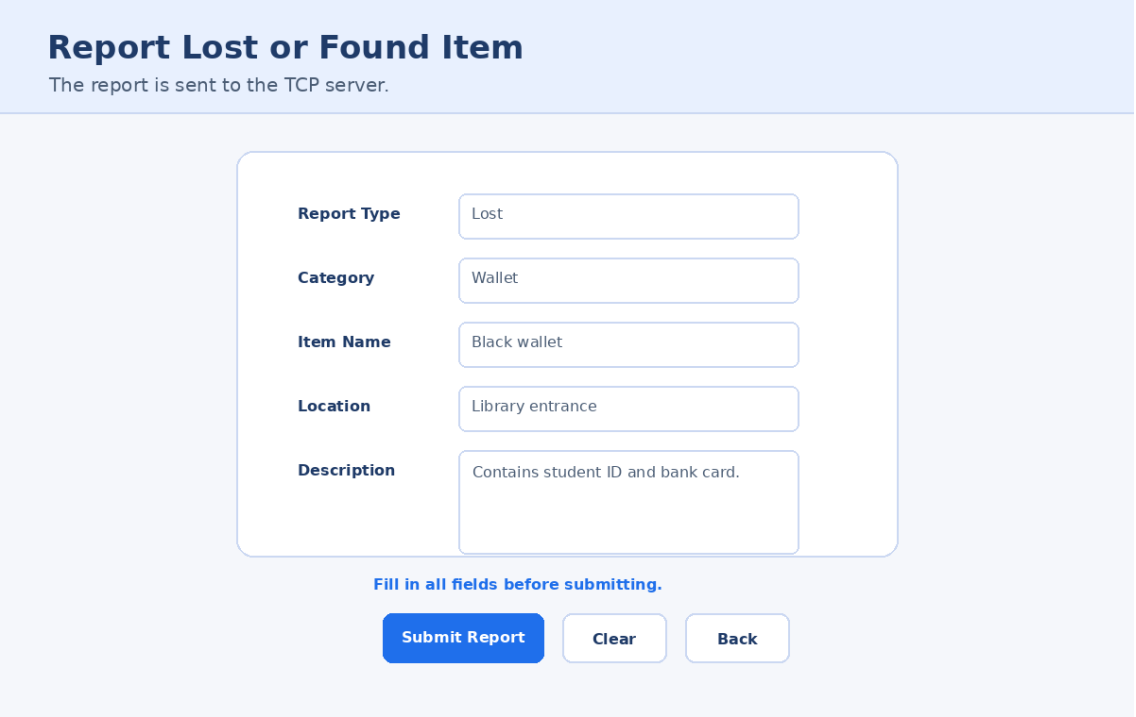
Login screen



Student dashboard connected to server

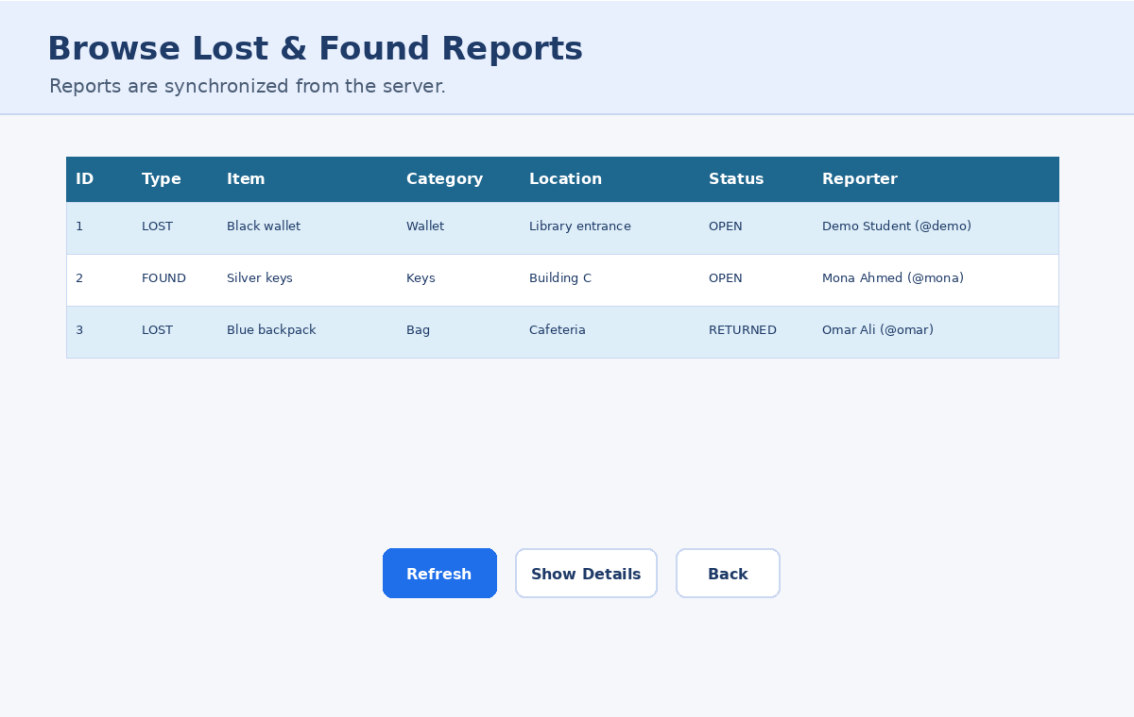


Report item screen

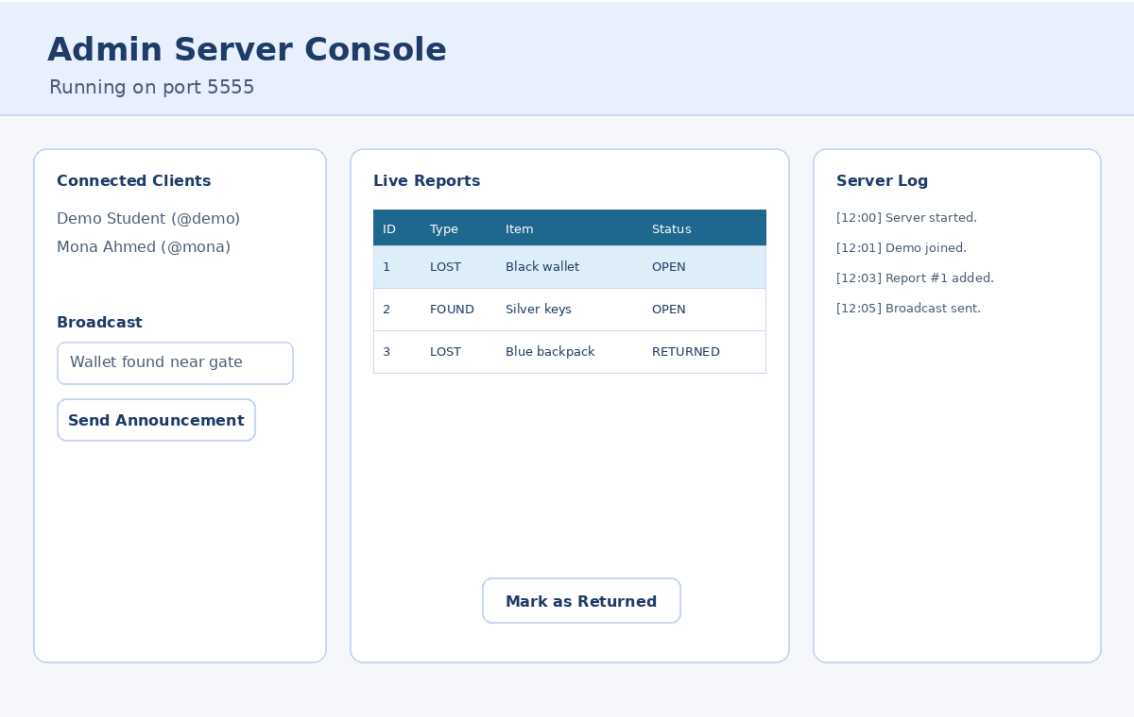


More Screenshots

Browse reports screen



Admin server console



6. Reflection

The hardest part was coordinating JavaFX with background networking threads without freezing the GUI. The solution was to keep socket communication in background threads and update UI state through observable lists and JavaFX-safe controller logic. If improved later, the project could add database persistence, item photo upload, and stronger authentication.

7. Partner Contribution Template

Partner 1: Client UI, login/register, dashboard, report form, README. Partner 2: Server networking, client handler threads, admin console, UML, and report. Replace this template with your real names before submission if required.